M7 · Summary & exam pearls

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

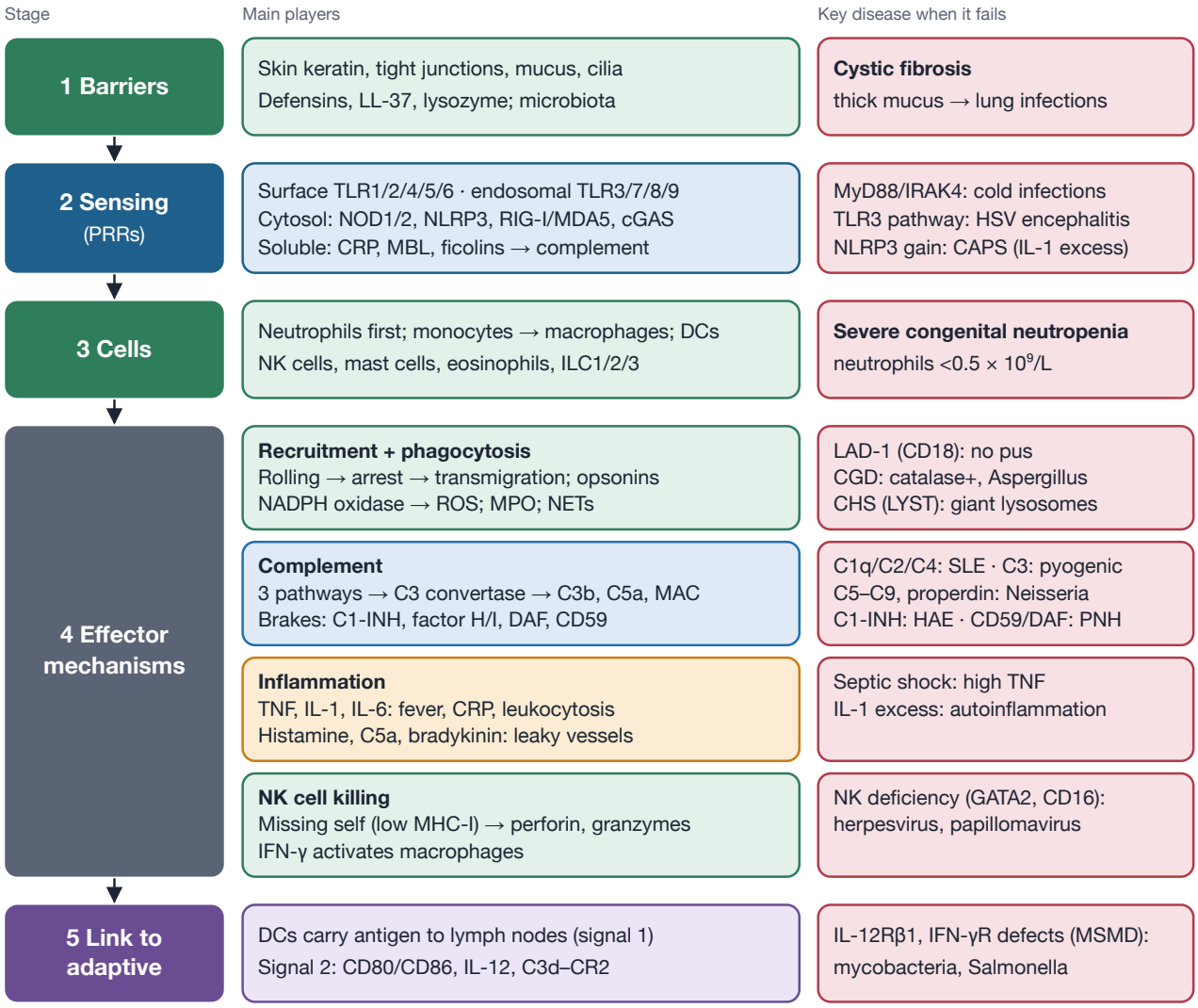
- Place each innate component on one map, from barrier to adaptive immunity.
- Recall the high-yield tables of M1–M6 in a few minutes.
- Match a clinical clue to the innate defect that causes it.
- Avoid the classic exam traps of innate immunity, complement and hereditary angioedema (HAE).

This module adds no new facts. Each statement repeats a fact from M1–M6, with the same fact ID and source page. Use it for rapid review. Go back to the module for detail.

PRR	pattern recognition receptor
TLR	Toll-like receptor
NK cell	natural killer cell
ILC	innate lymphoid cell
DC / pDC	dendritic cell / plasmacytoid DC
NET	neutrophil extracellular trap
CGD	chronic granulomatous disease
LAD	leukocyte adhesion deficiency
CHS	Chédiak-Higashi syndrome
MSMD	Mendelian susceptibility to mycobacterial disease
MPO	myeloperoxidase
MAC	membrane attack complex
C1-INH	C1 inhibitor
HAE	hereditary angioedema (HAE-1/2; HAE nC1-INH = HAE with normal C1-INH)
AAE-C1-INH	acquired angioedema with low C1-INH
CAPS	cryopyrin-associated periodic syndromes
PNH / aHUS	paroxysmal nocturnal haemoglobinuria / atypical haemolytic uraemic syndrome
SLE	systemic lupus erythematosus

1. The master map

Read the map from top to bottom. Each stage has a typical disease when it fails. The rule from M6 applies to all stages: the lost function predicts the disease.



D21. Innate immunity master map. Left: the five stages. Middle: the main players. Right: the key disease when that stage fails. Sources: barriers Abbas p.80; Janeway p.44 Abbas p.80; Janeway p.45, 44; sensing Abbas p.69, 72, 75, 76, 87, 88, 89 Janeway p.535; Abbas p.489, 75; cells Abbas p.94, 15-16, 20, 81 Janeway p.533; phagocytosis Abbas p.49, 96; Janeway p.49 Abbas p.488, 487; complement Abbas p.301, 307, 313, 311 Abbas p.315, 311; Janeway p.59; inflammation Abbas p.90-91, 96, 97, 94; Janeway p.84 Abbas p.97, 74; NK cells Abbas p.85, 82 Abbas p.488; link to adaptive Abbas p.100, 20, 93, 314; Janeway p.96 Abbas p.93, 498.

2. High-yield tables

2.1 Innate cells in one line

Cell	One line	Exam hook
Neutrophil	Most abundant white blood cell; main cell of acute inflammation; arrives first. Abbas p.14, 94	Azurophilic granules hold MPO, defensins and cathelicidins; NETs. Abbas p.14, 15
Macrophage	Most tissue-resident macrophages come from fetal precursors and self-renew. Abbas p.15	M1 kills microbes; M2 repairs tissue. SIRPα–CD47 is the "don't eat me" signal. Abbas p.19
Dendritic cell	Carries antigen to lymph nodes and links innate and adaptive immunity. Abbas p.20	pDC is the major type I IFN source; cDC1 cross-presents to CD8+ T cells. Abbas p.22, 21-23
NK cell	CD56+ CD3–; kills infected cells without prior activation. Abbas p.82	Low MHC-I ("missing self") removes inhibition; CD16 binds IgG for ADCC. Abbas p.85, 84
Mast cell	Not in blood; IgE cross-linking on FcεRI activates it. Abbas p.465, 468	Tryptase marks mast cell activation; C3a, C5a and MRGPRX2 act without IgE. Abbas p.471, 470
Basophil	Blood granulocyte (≤0.5%) with FcεRI; growth factor IL-3. Abbas p.465, 466	Matures in bone marrow, from a progenitor different from the mast cell progenitor. Abbas p.20
Eosinophil	IL-5 drives production; granules hold MBP and ECP. Abbas p.473	Low FcεRI without a signalling chain; eotaxin binds CCR3. Abbas p.473
ILCs	No TCR. ILC1: IFN-γ. ILC2: IL-5, IL-13 (GATA3). ILC3: IL-22, IL-17. Abbas p.81	IL-25, IL-33 and TSLP activate ILC2s, which can start allergic inflammation early. Abbas p.81, 465

2.2 PRR → ligand → location

PRR	Ligand	Location / note
TLR4	LPS (with LBP, MD2 and CD14)	Plasma membrane; uses MyD88 and TRIF Abbas p.67, 69
TLR1:TLR2, TLR2:TLR6	Peptidoglycan, lipoteichoic acid; tri-/diacyl lipopeptides	Plasma membrane Abbas p.67, 69; Janeway p.87
TLR5	Flagellin	Plasma membrane Abbas p.67
TLR3	dsRNA	Endosome; TRIF only Abbas p.67, 69
TLR7, TLR8	Viral ssRNA	Endosome Abbas p.67
TLR9	Unmethylated CpG DNA	Endosome; pDCs, B cells Abbas p.67; Janeway p.86
NOD1 / NOD2	DAP / muramyl dipeptide (peptidoglycan)	Cytosol → RIP2 → NF-κB Abbas p.72
NLRP3	Urate and cholesterol crystals, alum, ATP; K+ efflux, ROS	Cytosolic inflammasome → caspase-1 Abbas p.74, 72
RIG-I / MDA5	5′-triphosphate RNA / long dsRNA	Cytosol → MAVS → type I IFN Abbas p.75-76, 75, 76
cGAS → STING	Cytosolic dsDNA	STING in ER → TBK1 → type I IFN Abbas p.76
Dectin-1	Fungal β-glucan	SYK–CARD9; antifungal Th17 Abbas p.78
Mannose receptor	Terminal mannose, fucose, GlcNAc	Macrophages; phagocytosis Abbas p.78
FPR1	N-formylmethionyl peptides	Leukocytes; directs cell movement Abbas p.79
CRP / MBL (soluble)	Phosphorylcholine / mannose, fucose	Blood; C1q (classical) / MASP (lectin) Abbas p.87, 88, 86

2.3 Cytokine → main effect

Cytokine	Main effect
TNF	Activates endothelium and neutrophils; fever; acute-phase proteins; mediates septic shock. Abbas p.90, 92
IL-1β	Endothelium, fever, acute-phase proteins, Th17. Needs the inflammasome. Abbas p.90, 72
IL-6	Acute-phase reactants in the liver (CRP up to 1000-fold); Th17. Abbas p.93, 88
IL-12	IFN-γ from NK cells, ILC1s and T cells; Th1. Abbas p.93
IL-18	Inflammasome-dependent; enhances NK cell functions. Abbas p.94
Type I IFN	Antiviral state, more MHC-I, NK activation; pDCs make most IFN-α. Abbas p.90, 98
IFN-γ (NK cells)	Activates macrophages to destroy ingested microbes. Abbas p.82
IL-25, IL-33, TSLP	Epithelial; stimulate ILC2s, Th2 cells and mast cells. IL-33 is an alarmin. Abbas p.94
IL-10	Inhibits macrophages and DCs; IL-10 receptor loss causes severe infantile colitis. Abbas p.101
CXCL8 / CCL2	Attract neutrophils / monocytes. Abbas p.94

2.4 Phagocyte defect → clue

Defect (gene)	Step blocked	Clue
SCN (ELA2, HAX1); cyclic neutropenia (ELA2)	Production	Neutrophils <0.5 × 10^9/L; cyclic: about every 21 days. Janeway p.533, 534
LAD-1 (CD18)	Firm adhesion (β2 integrins)	Leukocytosis, no pus, poor wound healing. Abbas p.488
LAD-2 (GDP-fucose transporter)	Rolling (no sialyl Lewis X)	Developmental abnormalities; Bombay blood group. Abbas p.488
LAD-3 (KINDLIN-3)	Inside-out integrin activation	Bleeding with a normal platelet count. Abbas p.488
CGD (gp91 X-linked in about two-thirds; others AR)	NADPH oxidase (superoxide)	Catalase-producing bacteria (*Staphylococcus*), *Aspergillus* (leading cause of death), granulomas. Abbas p.486, 487
MPO deficiency	H_2O_2 → hypohalous acid	CGD-like, less severe. Janeway p.536
CHS (LYST)	Phagosome–lysosome fusion	Partial albinism, giant lysosomes, NK defect, HLH. Abbas p.488; Janeway p.536
MSMD (IL-12p40, IL-12Rβ1, IFN-γR, STAT1, NEMO)	IL-12/IFN-γ axis	Weak mycobacteria, *Salmonella*; BCG disseminates. Abbas p.498

2.5 Complement deficiency → phenotype

Missing protein	Phenotype
C1q, C2, C4	SLE in more than 50%. C2 deficiency is the most common. Abbas p.315
C3	Severe, possibly fatal, pyogenic infections. Abbas p.315
Factor I	Secondary C3 depletion → pyogenic infections. Abbas p.315
MBL, MASP2	Mild; bacterial infections in early childhood. Janeway p.533
Factor B, factor D, properdin (X-linked)	Meningococcal infections. Abbas p.315
C5–C9	Disseminated *Neisseria* only. Abbas p.315
Factor H	C3 consumption, glomerulonephritis; also aHUS (with MCP). Abbas p.315
CD59 + DAF (somatic PIG-A)	PNH: intravascular haemolysis, venous thrombosis. Abbas p.315, 312
C1-INH	HAE (autosomal dominant). Abbas p.311

2.6 Angioedema: types and lab pattern

Type	C1-INH function	C1-INH protein	C4	C1q	Clue
HAE-1 (about 85%)	↓	↓	usually ↓	normal	Onset in childhood; family history absent in up to 25%. HAE-GL p.5, 6, 7, 10
HAE-2	↓	normal or ↑	usually ↓	normal	As HAE-1. HAE-GL p.5, 6
AAE-C1-INH	↓	↓	↓	↓ in 75%	Later onset; lymphoma or MGUS; autoantibodies to C1-INH. HAE-GL p.7
HAE nC1-INH	normal	normal	Not covered in provided sources — needs expert input.		Very rare; FXII, ANGPT1, PLG or unknown mutation. HAE-GL p.6
ACE inhibitor-induced	normal	normal	—		Acquired, bradykinin-mediated. HAE-GL p.6

Test every suspected patient for C1-INH function, C1-INH protein and C4. Repeat low results. HAE-GL p.6 C3 and CH50 are normal in HAE. HAE-GL p.6-7 Measure C1q in new angioedema after age 40 without a family history. HAE-GL p.7

2.7 HAE treatment (2017 guideline — volatile)

> **Check the current guideline**
>
> This table follows the WAO/EAACI HAE guideline, **2017 revision**, and applies to HAE-1/2 only. HAE-GL p.6 A 2021 revision exists and is not in the sources. Check every drug, licence and line of therapy against the current guideline.

Setting	2017 recommendation
On-demand (attack)	Treat early. Use C1-INH, ecallantide or icatibant; if not available, SDP, then FFP. HAE-GL p.7, 7-8 Do not use tranexamic acid or androgens. HAE-GL p.8
Supply	Every patient carries on-demand medication for two attacks. HAE-GL p.8
Short-term prophylaxis	Before procedures: C1-INH concentrate close to the procedure. HAE-GL p.9
Long-term prophylaxis	First line: C1-INH. Second line: androgens. Antifibrinolytics are not recommended. HAE-GL p.9-10, 10
Children <12 years	C1-INH for attacks. HAE-GL p.11
Pregnancy, lactation	C1-INH is preferred; androgens are contraindicated. HAE-GL p.13

3. Classic exam traps

1. TLR3 is the only TLR that does not use MyD88. TLR4 is on the surface but uses both MyD88 and TRIF. Abbas p.69
2. Pyroptosis affects macrophages and DCs, not neutrophils. Abbas p.74
3. Weak NOD2 variants increase Crohn's disease risk; NOD2 gain of function causes Blau syndrome. Abbas p.72
4. Tryptase, not histamine, is the marker of mast cell activation. Abbas p.471
5. Eosinophils have low FcεRI without a signalling chain. There is no convincing evidence that IgE activates them. Abbas p.473
6. Lysozyme works better against Gram-positive bacteria. LPS covers the peptidoglycan of Gram-negative bacteria. Janeway p.45
7. Anti-TNF drugs gave no benefit in sepsis trials and can reactivate tuberculosis. Abbas p.97; Janeway p.109
8. In LAD, blood shows leukocytosis, but tissue has no pus. Abbas p.488
9. NADPH oxidase makes superoxide. MPO makes hypohalous acid from H_2O_2. Abbas p.96
10. For C2, the larger bound fragment is C2a. Abbas p.305
11. Free IgG or IgM does not activate C1; IgG4 does not activate complement. Abbas p.305
12. Factor H regulates only the alternative pathway. C4BP regulates C4b. Abbas p.311, 310
13. CD59 blocks C9 at C5b-8; DAF dissociates the C3 convertases. Abbas p.312, 310
14. Normal C1-INH does not exclude HAE (HAE nC1-INH). HAE-GL p.6

15. Wheals do not exclude HAE, because chronic spontaneous urticaria can coexist. Erythema marginatum is often mistaken for urticaria. HAE-GL p.7, 11
16. C4 is not useful for HAE diagnosis below age 12 months. HAE-GL p.11
17. PNH is not inherited. The PIG-A mutation is somatic. Abbas p.312

4. If you see X, think Y

If you see …	… think of
Early invasive pneumococcal infection, no fever, little inflammation	MyD88 or IRAK4 deficiency Abbas p.489; Janeway p.535
Recurrent herpes simplex encephalitis, other viral defense normal	TLR3 pathway defect (TLR3, UNC93B, TRIF) Abbas p.489
Cold-induced fever, urticarial rash, dominant inheritance	FCAS (NLRP3); with deafness: Muckle-Wells Janeway p.537
Severe herpesvirus and papillomavirus infections	NK cell deficiency (e.g., GATA2, CD16) Abbas p.488
Staphylococcus abscesses, *Aspergillus*, granulomas	CGD Abbas p.487
Infections without pus, leukocytosis, poor wound healing	LAD-1 Abbas p.488
Infections plus bleeding, normal platelet count	LAD-3 Abbas p.488
Partial albinism, silver hair, giant lysosomes	Chédiak-Higashi syndrome Abbas p.488; Janeway p.540
Disseminated BCG or *Salmonella*	MSMD (IL-12/IFN-γ axis) Abbas p.498
Recurrent disseminated *Neisseria*	C5–C9, properdin or factor D deficiency Abbas p.315; Janeway p.532
Lupus-like disease with a complement deficiency	C1q, C2 or C4 deficiency Abbas p.315
Low C3, glomerulonephritis, autoantibody stabilizing C3bBb	C3 nephritic factor Abbas p.315
Child with haemolytic anaemia, thrombocytopenia, acute renal failure	aHUS (factor H, MCP) Abbas p.315
Intravascular haemolysis and venous thrombosis	PNH (no CD59, DAF) Abbas p.312, 315
Recurrent swelling, abdominal attacks, no wheals, no response to antihistamines	HAE-1/2 HAE-GL p.6
New angioedema after 40, no family history, low C1q	AAE-C1-INH (lymphoma, MGUS) HAE-GL p.7
Angioedema on an ACE inhibitor, normal C1-INH	ACE inhibitor-induced angioedema HAE-GL p.6
Raised serum tryptase	Mast cell activation (anaphylaxis); mastocytosis Abbas p.471, 470-471
Charcot-Leyden crystals	Eosinophil galectin-10 Abbas p.473

5 key points

1. Innate immunity uses germline-encoded, nonclonal receptors and has limited memory. Abbas p.63, 64, 4
2. Location predicts the ligand: surface TLRs see microbial walls; endosomal TLRs see nucleic acids. Abbas p.69
3. The lost function predicts the disease: early classical → SLE; C3 → pyogenic; C5–C9 → *Neisseria*. Abbas p.315
4. HAE-1/2 is bradykinin-mediated. Test C1-INH function, C1-INH protein and C4; use C1q for acquired deficiency. HAE-GL p.6, 7
5. Innate immunity gives signal 2 (costimulators, cytokines, complement products) and starts adaptive immunity. Abbas p.100

3 exam traps

1. TLR3 is the only TLR without MyD88; TLR4 uses MyD88 and TRIF. Abbas p.69
2. C3 and CH50 are normal in HAE; normal C1-INH does not exclude HAE. HAE-GL p.6-7, 6
3. LAD gives leukocytosis in blood but no pus in tissue. Do not confuse it with neutropenia. Abbas p.488

Practice: the whole quiz (all modules M1–M6 and the SCT samples). One-page review: exam pearls.

Sources: Abbas, Cellular and Molecular Immunology, 11th ed. (pp. 3–101, 297–316, 465–498); Janeway's Immunobiology, 8th ed. (pp. 37–111, 532–540); WAO/EAACI HAE guideline, 2017 revision (pp. 5–15). This module reuses fact IDs from the M1–M6 ledgers only · Fact IDs refer to `ledger/facts.csv`.